

Documentation

Why S3?

Simplicity: S3 is a straightforward and cost-effective solution for hosting static websites. It eliminates the need for managing servers or complex configurations.

Scalability: S3 can handle high traffic loads without any additional setup.

Dynamic Content: By using a placeholder (`${var.dynamic_string}`) in the HTML file, the content can be updated dynamically without redeploying the infrastructure.

How to Update the Dynamic String

Update the `dynamic_string` variable in the Terraform configuration.

Run `terraform apply` to update the S3 bucket object with the new string.

Future Enhancements

API Gateway + Lambda: To make the dynamic string truly dynamic (without requiring a `terraform apply`), you could use an API Gateway with a Lambda function. The Lambda function would fetch the string from a database (e.g., DynamoDB) and return it to the frontend via an API call.

CI/CD Pipeline: Integrate a CI/CD pipeline (e.g., GitHub Actions, Jenkins) to automate the deployment process whenever the HTML content or dynamic string changes.

CloudFront: Add a CloudFront distribution in front of the S3 bucket to improve performance and reduce latency for global users.

Environment Variables: Use environment variables or secrets management (e.g., AWS Secrets Manager) to store sensitive data like API keys or database credentials.

Monitoring and Logging: Implement CloudWatch alarms and logs to monitor the health and performance of the infrastructure.

How to Embellish the Solution

Dynamic Updates via API: Replace the S3-hosted HTML with a frontend framework (e.g., React) that fetches the dynamic string from an API endpoint. This would allow real-time updates without redeploying the infrastructure.

Database Integration: Store the dynamic string in a database (e.g., DynamoDB) and use a Lambda function to fetch and update it. This would make the solution more robust and scalable.

User Authentication: Add user authentication (e.g., Cognito) to restrict access to the page or allow users to update the dynamic string.

<http://my-dynamic-string-bucket.s3-website-us-east-1.amazonaws.com>